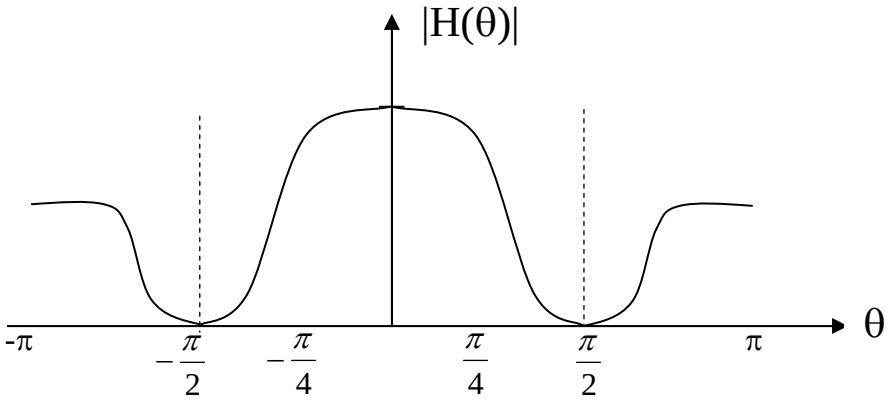
Final Examination, Session 1, 2006
Model Answers

Question 1
(A)

$$H(z) = K\ \frac{(z - j)(z - j)}{(z\ - 0.5j)(z + 0.5j)} = K\frac{\left(z^2 + 1\right)}{z^2 + \frac{1}{4}}$$

$$= K\ \frac{(1 + z^{-2})}{1 + 0.25z^{-2}}$$



(B)

(i)

$$H(z) = a_0 + \frac{1}{3}z^{-1} + a_2 z^{-2}$$

$$H(\theta)\big|_{\theta=0} = 1 = a_0 + \frac{1}{3} + a_2$$

For linear phase $a_0 = a_2$

$$1 = a_0 + \frac{1}{3} + a_0$$

$$\therefore a_0 = \frac{1}{3}, \quad a_2 = \frac{1}{3}$$

(C)

(i)

$$R(z) = X(z) + z^{-1}Y(z)(-r\sin\theta_0) + r\cos\theta_0 R(z)z^{-1}$$

$$R(z)\left(1 - r\cos\theta_0 z^{-1}\right) = X(z) - r\sin\theta_0 z^{-1}Y(z) \quad ----(1)$$

$$Y(z) = R(z)r\sin\theta_0 z^{-1} + Y(z)r\cos\theta_0 z^{-1}$$

$$R(z) = \frac{Y(z)\left[1 - r\cos\theta_0 z^{-1}\right]}{r\sin\theta_0 z^{-1}} \quad -----(2)$$

From (1) and (2):

$$\frac{X(z) - r\sin\theta_0 z^{-1}Y(z)}{1 - r\cos\theta_0 z^{-1}} = \frac{Y(z)\left[1 - r\cos\theta_0 z^{-1}\right]}{r\sin\theta_0 z^{-1}}$$

$$X(z)r\sin\theta_0 z^{-1} - r^2\sin^2\theta_0 z^{-2}Y(z) = Y(z)\left[1 - 2r\cos\theta_0 z^{-1} + r^2\cos^2\theta_0 z^{-2}\right]$$

$$\therefore \frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2\theta_0 z^{-2}}$$

(ii)

$$\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2\theta_0 z^{-2}}$$

D

(i)

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

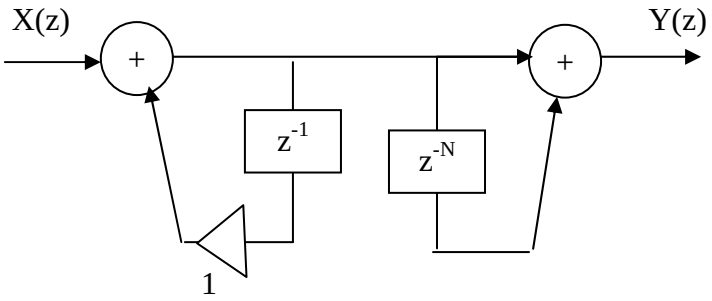
$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j5\theta}}{1 - e^{-j\theta}} = \frac{e^{-j\frac{5}{2}\theta}}{e^{-j\frac{1}{2}\theta}} \frac{\left(e^{j\frac{5}{2}\theta} - e^{-j\frac{5}{2}\theta}\right)}{\left(e^{-j\frac{1}{2}\theta} - e^{-j\frac{1}{2}\theta}\right)}$$

$$H(\theta) = e^{-j2\theta}\frac{\sin\frac{5}{2}\theta}{\sin\frac{1}{2}\theta}$$

(ii)

$$H(z)=\frac{1-z^{-5}}{1-z^{-1}}$$



(iii)

$$\phi(\theta)=-2\theta$$

$$\therefore \tau_g=-\frac{d\phi(\theta)}{d\theta}=2 \quad \text{(constant group delay)}$$

(E)

$$H(z)=\left(1-z^{-1}\right)^3$$

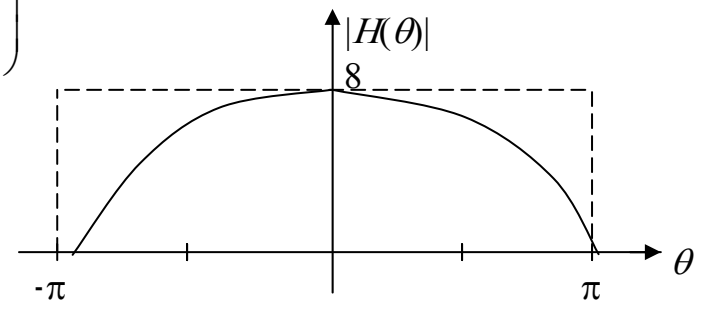
$$H(\theta)=\left(1-e^{-j\theta}\right)^3$$

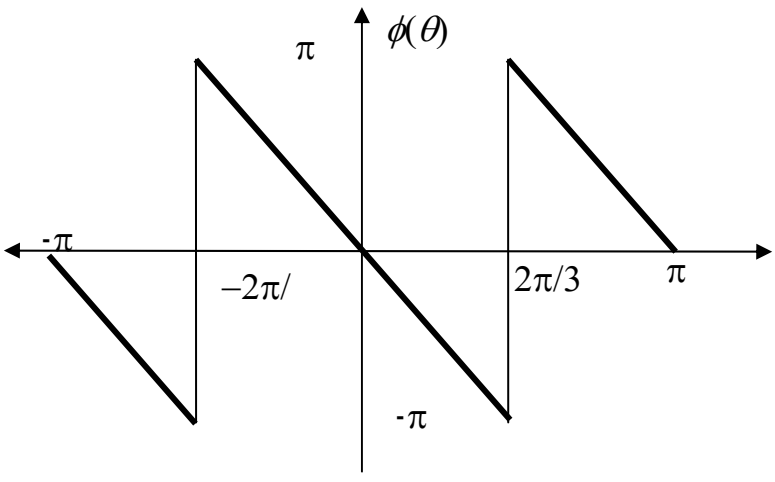
$$=\left(-e^{-j\frac{\theta}{2}}\right)^3\left(e^{j\frac{\theta}{2}}+e^{-j\frac{\theta}{2}}\right)^3$$

$$=e^{-j\frac{3\theta}{2}}\times 8\times\cos^3\tfrac{\theta}{2}$$

$$\therefore |H(\theta)|=\left|8\cos^3\tfrac{\theta}{2}\right|$$

$$\therefore \phi(\theta)=-\tfrac{3}{2}\theta$$





(F)

$$H(\theta)=\frac{k_0^2}{\left(1-b_1e^{-j\theta}\right)^2}$$

$$H(\theta)\big|_{\theta=0}=1 \quad \Rightarrow \quad 1=\frac{k_0^2}{(1-b_1)^2} \quad \Rightarrow k_0^2=(1-b_1)^2$$

$$|H(\theta)|^2=\frac{k_0^2}{\left(1+b_1^2-2b\cos\theta\right)^2}$$

$$|H(\theta)|^2_{\theta=\frac{\pi}{4}}=\tfrac{1}{4}=\frac{k_0^2}{\left(1+b_1^2-2b\times\frac{1}{2}\right)^2}$$

$$\therefore 1+b_1^2-b_1=k_0\,2=2(1-b_1)$$

$$b_1^2+b_1-1=0$$

Low pass filter $\quad b_1=0.62$

$$\therefore k=1-b_1=0.38$$

Question 2

(A)

(i)

From the 1st equation:

$$\cos(n+1)\theta=2\cos\theta+\cos n\theta-\cos(n-1)\theta$$

From the 2nd equation:

$$\sin n\theta=\frac{\cos(n+1)\theta-\cos(n-1)\theta}{-2\sin\theta}$$

Let $\begin{aligned} y_1(n)&=\cos n\theta \\ y_2(n)&=\sin n\theta \end{aligned}$

$$y_1(n+1)=2\cos\theta y_1(n)-y_1(n-1) \quad ----(1)$$

$$y_2(n)=\frac{1}{2\sin\theta}\left[y_1(n-1)-y_1(n+1)\right] \quad ----(2)$$

(ii) Draw the block diagram from the above calculation.

(iii)

$\theta = \frac{3}{4}\pi$

$\therefore a = 2\cos\frac{3}{4}\pi = - = \sqrt{2}$

$$b = \frac{1}{2\sin\frac{3}{4}\pi} = \frac{\sqrt{2}}{2}$$

(B)

$$\theta_1 = 2\pi \times 2 \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta - \theta_1) + X(\theta + \theta_1)\right] = \frac{1}{2}\left[1 + \cos\left(\theta - \frac{\pi}{2}\right) + 1 + \cos\left(\theta + \frac{\pi}{2}\right)\right] = \frac{1}{2}\times 2 = 1$$

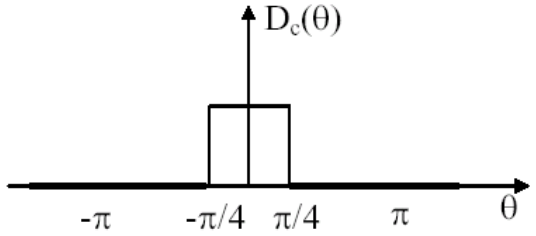
$$D_C(\theta) = B_C(\theta)\cdot G(\theta) = 1, \quad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta) = \frac{1}{2}\left[D_C(\theta - \theta_2) + D_C(\theta + \theta_2)\right] = \frac{1}{2}\left[D_C\left(\theta - \tfrac{3}{4}\pi\right) + D_C\left(\theta + \tfrac{3}{4}\pi\right)\right]$$

$$R_C(\theta) = \begin{cases} \frac{1}{2} & -\frac{\pi}{4} \le \theta - \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\frac{\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow -\pi \le \theta \le -\frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta),$ and $R_C(\theta)$.

$B_c(\theta) = 1, D_c(\theta) = 1$

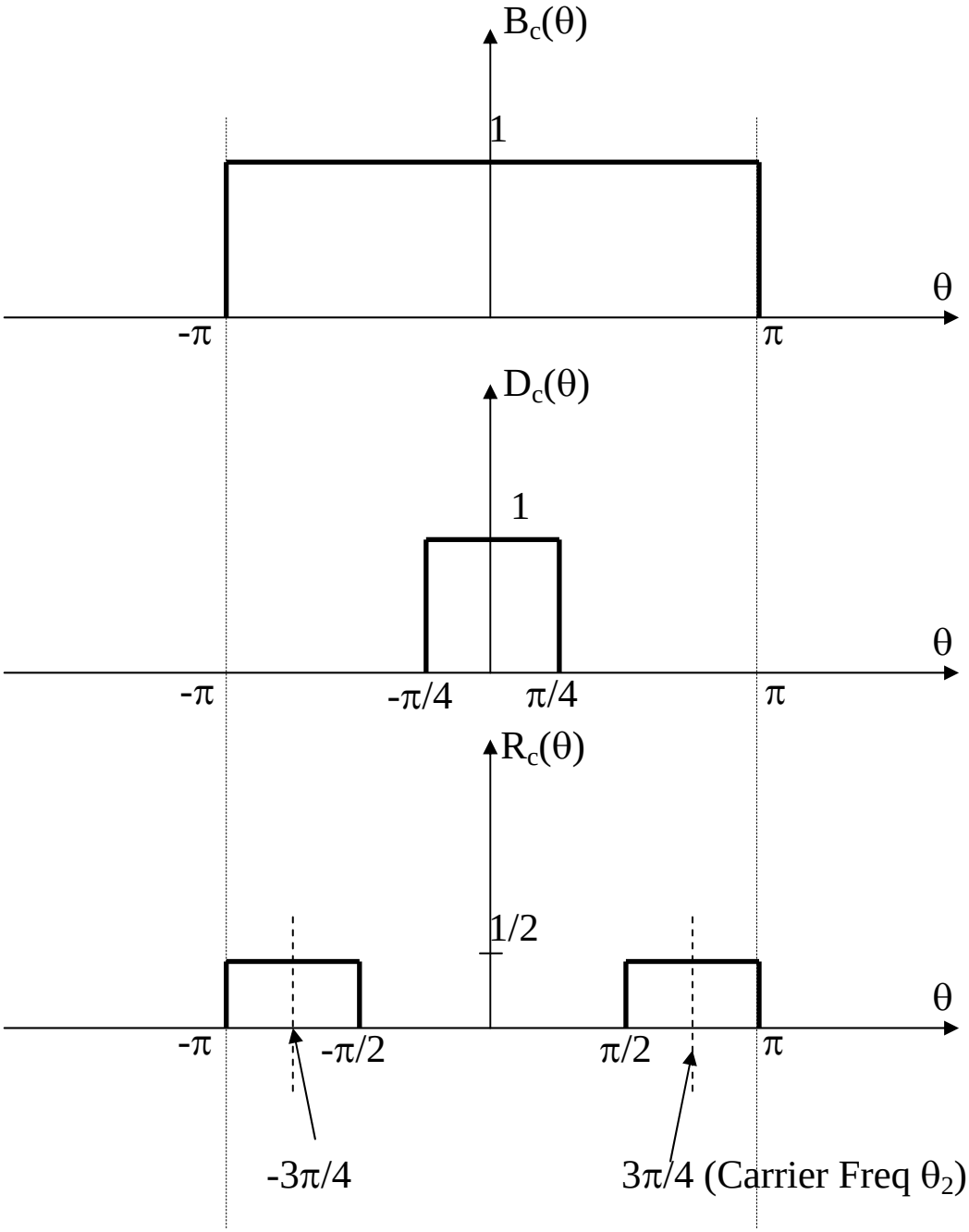


$$R_c(\theta) = \frac{1}{2}\{D_c(\theta - \theta_1) + D_c(\theta + \theta_1)\}$$

$$= \frac{1}{2}\left\{D_C(\theta - \frac{3\pi}{4}) + D_c(\theta + \frac{3\pi}{4})\right\}$$

$$= \begin{cases} \frac{1}{2} & \frac{-\pi}{4} \le (\theta - \frac{3\pi}{4}) \le \frac{\pi}{4} \\ \frac{1}{2} & \frac{-\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\pi \le \theta \le \frac{-\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$


$B_c(\theta)$
1
θ
-π
π
$D_c(\theta)$
1
θ
-π
-π/4
π/4
π
$R_c(\theta)$
1/2
θ
-π
-π/2
π/2
π
-3π/4
3π/4 (Carrier Freq θ_2)

Question 3:
(A)

$$H(s)=\frac{0.1s}{s^2+0.1s+1}$$

$$\theta_C=2\pi\frac{fc}{fs}=\frac{\pi}{2}$$

$$\therefore \omega'_C=\frac{2}{T}\tan\left(\frac{\theta_C}{2}\right)=\frac{2}{T}\ \text{rad/sec}$$

Pre-warped analogue filter:

$$H(s)=\frac{0.1\times\frac{s}{\frac{2}{T}}}{\left(\frac{s}{\frac{2}{T}}\right)^2+0.1\times\frac{s}{\frac{2}{T}}+1}$$

$$H(s)\big|_{s=\frac{2}{T}\frac{z-1}{z+1}}=H(z)=\frac{0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}}{\left(\frac{2}{T}\right)^2\left(\frac{z-1}{z+1}\right)^2+0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}+\left(\frac{2}{T}\right)^2}$$

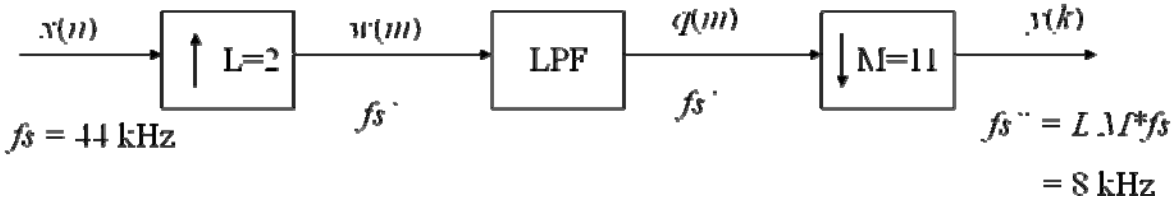
$$H(z)=\frac{1}{21}\frac{z^2-1}{z^2+\frac{19}{21}}=\frac{1}{21}\frac{1-z^{-2}}{1+\frac{19}{21}z^{-2}}$$

(B)

$$h(n)\xrightarrow{DTFT}H(\theta)$$
$$v(n)=x(-n)*h(n)$$
$$u(n)=v(-n)=x(n)*h(n)$$
$$y(n)=\{h(n)+h(-n)\}*x(n)$$
$$Y(z)=\left\{H(z)+H\left(z^{-1}\right)\right\}\bullet X(z)$$
$$G(z)=H(z)+H\left(z^{-1}\right)$$
$$G(\theta)=H(\theta)+H^*(\theta)=(a+jb)+(a-jb)=2a$$

Hence the equivalent transfer function $G(\theta)$ is real and has zero phase.

(C)
(i)



$$\frac{L}{M} = \frac{8000}{44000} = \frac{2}{11}$$

(ii)
Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.